

**EEHC DISTRIBUTION MATERIALS SPECIFICATION
BATTERY CHARGER and DC SOURCE 24 V/ 30 ADC**

**EDMS 27-304-1
24-10-2023**

**EDMS 27-304-1
TECHNICAL SPECIFICATION
FOR
BATTERY
CHARGER and DC SOURCE 24 V/ 30 ADC**

Issue: Jan -2023/ Rev- 1

توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

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**EEHC DISTRIBUTION MATERIALS SPECIFICATION
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1. SCOPE:

This specification covers the minimum technical requirements for design, engineering, manufacture, inspection, testing and performance of indoor battery chargers intended to be used in the primary substation of Egyptian Electricity Distribution Company.

2. APPLICABLE STANDERDS:

Unless otherwise specified in this specification, the indoor battery chargers should comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance with the applicable IEC standards or approved equivalent including but not limited to the following:

IEC 61000-6-2	Electromagnetic compatibility (EMC) – Part 6-2: Generic standards – Immunity for industrial environments
IEC 61000-6-4	Electromagnetic compatibility (EMC) – Part 6-4: Generic standards – Emission standard for industrial environments
IEC 60146-1-1	Semi-Conductor Converters- General Requirements and line commutated Converters-Part-1-1. Specification of basic requirements.
IEC 60146-2	Semiconductor converters - Part 2: Self-commutated semiconductor converters including direct d.c. converters
IEC 62040-1	Power systems (UPS) – Part 1: General and safety requirements for UPS.
IEC 62040-3	Uninterruptible power systems (UPS) – Part 3: Method of specifying the performance and test requirements
IEC 60076 / EN 60076	Power transformer and reactor cooling equipment.
EN 61558	Safety of transformers, reactors, power supply units and combinations thereof
ANSI/ NEMA ICS6	Enclosures for Industrial Controls and Systems
NEMA PE5	Constant-Potential-Type Electric Utility (Semi-Conductor Static Converter) Battery Chargers
NEMA 250	Enclosures for Electrical Equipment (1000 V maximum)
IEC 60529	Classification of Degrees of Protection Provided by Enclosure.
ANSI C63.4	Measurement of Radio Noise Emission From low Voltage Electrical and Electronic Equipment in the Range of 9 KHZ to 40 GHZ.
IEEE 446	Recommended Practice for Emergency and Standby Power System for Industrial and Commercial Application.

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3. ENVIRONMENTAL CONDITIONS:

The performance of indoor battery chargers should be guaranteed for the following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Minimum ambient temperature	-5°C
Maximum ambient	45°C (50 °C as option).
Storage temperature	-20 to + 70 °C
Maximum relative humidity	100 % non-condensing.
Protection degree	IP: 42
Cooling	Natural and forced air ventilation.
Efficiency at full load	≥ 85 %.
Noise level	Max. 48 dB.
Maximum altitude	1000 m

4. DESIGN AND CONSTRUCTION REQUIREMENTS:
4.1 GENERAL

- 4.1.1 Power conversion: Phase rectification using thyristors and diodes in full wave bridge connection
- 4.1.2 The battery charger shall be constant voltage type solid state (Silicon) controlled rectifier equipment, designed to provide fully automatic voltage stabilization and current limitation for charging batteries in DC Systems. It shall also be capable of simultaneously supplying varying load permanently connected in parallel with the battery.
- 4.1.3 The charger circuitry shall be of modular design and the printed circuit cards shall be plug-in or slide-in type. The silicon rectifier shall be full wave controlled via thyristors shall be assembled to one sub-assembly, which can be easily exchanged.
- 4.1.4 It shall be ensured that harmonics due to SCR (Silicon Controlled Rectifier) commutations are not reflected back into the AC power mains, according to IEC61000-3-12, R_{sc}=33.
- 4.1.5 Disconnection of DC load from the supply during high-rate charging is not permitted. It is necessary to install voltage-dropping heavy duty power diodes in the load circuit to ensure that the load voltage does not rise above its permissible upper limit. This requirement of dropping diodes is more considerate with the Nickel Cadmium batteries. The diodes shall be automatically switched in and out by means of voltage sensing relay and contactor on a fail-safe basis. This means that on failure of the switching relay and contactor the diodes shall be in circuit.
- 4.1.6 The charger shall have the followings
 - Earth copper bar for earth bonding (4 ways - 30mm × 3 mm).
 - Ventilation with IP 42.
 - Lock for door with key.
 - Metallic stand with height not less than 15 cm.

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- The base of charger shall equipped with glands for connecting external wires
- Hooks for lifting
- The charger enclosure thickness shall be not less than 2mm

4.2 PERFORMANCE CHARACTERISTICS AND RATINGS

- 4.2.1 The charger shall have float, high-rate and initial charging modes. The charger must have the possibility of both manual and automatic switching over from one regime to the other one depending on the state of battery current independently.
- 4.2.2 The charger rectifier bridge and main transformer shall be rated for the required charger rated output current of 30 A permanently at maximum ambient temperature, this output shall feed both battery charging and load which are connected in parallel at charger output.
- 4.2.3 The charger is to be equipped on all sides (input and output) with circuit breaker having thermal and magnetic releases on all poles and with the following specification:
- A.C supply side (A.C) suitable for protecting charger with output (2 pole MCB 32A – 10 kA).
 - Battery side (DC) suitable for max. Charging current of (2 pole MCB 30 A- 10 kA).
 - Load Main (DC) suitable for max. Charging current of (2 pole MCB 30 A- 10 kA).
 - RTU (DC) suitable for max.; load of (2 pole MCB 16 A- 10 kA).
 - Spare (DC) suitable for max.; load of (2 pole MCB 16 A- 10 kA).
- 4.2.4 The charger shall have the following specifications:
- AC supply voltage:..... 220 V/AC $\pm$ 15%
 - Output DC voltage:
 - Float charging: 20 to 30 (to be set at 28 V)
 - High rate charging:..... 28 to 34 (to be set at 31 V)
 - Load voltage:..... $24 \pm 2\%$ at float and high-rate modes
 - rated current 30 A
 - Current limitation 0% to 100 % of rated current.
 - Input power factor:..... > 0.8
 - Long term stability:..... 0.15% of DC.
 - Ripple voltage:
 - Battery connected:..... < 1% Peak to Peak of VDC.
 - Battery not connected:..... < 2% rms of VDC.
 - Current Regulation:..... From 0 to 2% from current limit, and short circuit, with 10 to 100% setting range, as per IEC 60146.
 - Insulation:..... > 2 KV AC for one minute between input/output and earth.
 - Insulation resistance:..... > 10 M Ω with 500 VDC between input / output and earth.

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- Charging characteristic:..... Constant current / constant voltage with max. Current limitation (I/U as per DIN 41773-1 for LA batteries, DIN 41773-2 for NiCd batteries).
- Static voltage regulation:..... The static DC voltage must not deviate by more than 1% of the preset value while:
 - The AC voltage is simultaneously varying between 85% and 115% of UAC
 - The frequency within ± 6 Hz of the nominal frequency.
 - The load between 0-100 % of the Rated current range of the rectifier and the ambient temperature range.
- Long-term stability:..... Long-term stability 0.15 % of U_{rated} per 1000 hrs.
- Dynamic voltage regulation:..... The maximum permissible voltage surge occurring in the rectifier under floating charge conditions in the event of an instantaneous reduction in load from 100% to 10% of the rated current, or in the event of an instantaneous change in the supply voltage ranging from 0 to 110% of the rated value, is not be more than 5% or less than 0.5% after 4 seconds.
- The load regulating unit shall be via dropping diodes and to be capable to carry a load of 32A permanently at maximum ambient temperature.
- The output DC voltage of the charger shall not be affected by reversing of the phase sequence.
- Upon mains failure, the charger shall remain supplying the DC load from the battery without interruption and shall be capable to measure battery voltage and current. Also the communication module and the horn must remain in operation.
- The charger shall operate automatically after the cause of trip is removed.

4.3 CONTROL, PROTECTION, STATUS INDICATIONS, ALARM AND METERING
DEVICES

- 4.3.1 An automatic and adjustable current limiting device shall be provided to protect the charger against direct short circuits and continuous overloads without tripping it out of circuit. The setting range shall be up to 100% of the rated current of the charger. Normal function of the charger shall be restored automatically at the end of overloads/short circuits. The charger output current must be automatically limited to the specified value; up to 100% loss of battery voltage (e.g., 100% of cells are shorted).
- 4.3.2 The charger shall be protected against reverse feeding from any external source or battery.
- 4.3.3 Charger redundancy:
 - The charger must be suitable for parallel redundant operation with similar one as load sharing.

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- 4.3.4 The charger shall initiate the high-rate mode either manually by manual selector switch or automatically time adjustable up to 15 hours adjustable timer. The high rate charging mode, shall automatically return to float mode upon expiry of the high rate time
- 4.3.5 The initial charging mode, shall automatically return to float mode upon expiry of the initial time adjustable up to 15 hours acc. to IEC60623, this mode of operation must only be enabled; when the (MCBs) of the load and lighting are off.
- 4.3.6 The charger shall be furnished with a soft start to limit the inrush current upon application of input voltage. The control circuitry shall gradually ramp up the output current to allow gradual loading of the input AC power source.
- 4.3.7 The charger shall be equipped with protection, control, indication, alarm and metering devices.
- 4.3.7.1 Control
- On/off switch
 - Mode selector:
 - Manual: float, high rate, Initial
 - Automatic: float and high rate
 - Settings for all parameters:
 - Float charging voltage
 - High-rate charging voltage
 - Initial charging voltage
 - Charger, Battery and Load over currents.
 - Charger and Load Over and under voltages.
 - Current Limitation.
 - High rate Time.
 - Initial Time.
 - Battery End Voltage for Low Voltage Disconnect (LVD).
 - Alarm reset / horn stop.
 - Lamp Test.
- 4.3.7.2 Protections:
- The charger shall trip for specified value and shall operate again automatically when normal state is recovered.
- AC Input Side:
 - Phase failure (not phase sequence). (With trip, lamp, and horn)
 - Over voltage and under voltage. (With trip, lamp, and horn)
 - Input C.B for over load and short circuit protection.
 - Varistors (A.C surge suppression).
 - DC Output Side:
 - Reverse battery polarity connection.
 - Charger Over Voltage. (With trip, lamp, and horn)
 - Charger Under Voltage. (With trip, lamp, and horn)
 - Charger Over Current. (With trip, lamp, and horn)
 - Battery End Voltage. (With trip, lamp, and horn)

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- Load Regulating Unit (dropping diode) Failure. (With lamp, and horn).
- Adjustable LVD for battery disconnect at discharge. (With lamp, and horn)
- Short circuit and overload limitation.
- 4.3.9.3 Status Indications, Alarms and Instrumentations
- Alarm Actions (Trip, Horn and lamp) For A.C. And D.C shall be as described in section “4.3.9.2”:
- Charger failure (red lamp) with free contact for remote alarm
- A.C. failure Lamp
- Charger On (Green lamp)
- Any Circuit breaker trip with lamp.
- Light for each operating mode (Manual, Automatic, Float, High rate and Initial)

4.3.7.4 Metering and Measurements:

- The charger shall provide all the following measurements:
 - Battery and load DC voltages.
 - Phase AC voltages.
 - Charger output Current.

4.3.7.5 The horn alarm time should be adjustable; where it can be reseted after pre-set time.

4.3.7.6 The charger shall be equipped with the voltage free contact relays for the following alarms and functions:

- General charger failure
- DC under voltage
- AC Failure

4.4 COMMUNICATION

The charger shall communicate with the station RTU via Modbus TCP/IP communication.

Signal List

Item	Signal Description	Signal acquisition
Measurements	Phase input voltage	measured values/Comm.
	DC battery voltage	measured values/Comm.
	DC load voltage	measured values/Comm.
	Charger Output current	measured values/Comm.
Alarms	AC voltage Fault	Status values/Comm.
	DC Over/under voltage	Status values/Comm.
	Any MCB trip	Status values/Comm.
	earth fault	Status values/Comm.
	Charger failure	Hard wired

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5. INSPECTION:

.....EDC representatives should attend the factory tests according to section 6.2 and fill the attached Appendix 1/1 during the tests to ensure the charger complies with the EEHC specification.

6. TESTS:

All test results shall be provided for review and acceptance by ...EDC. The following table from IEC 60146 gives the required Type, Routine and Optional Test.

Test	Type Test	Routine Test	Optional Test	Specification subclause at IEC 60146-1-1
Visual inspection	X	X		
Insulation test	X	X		7.2
Light load and function test	X	X		7.3.1
Rated test	X			7.3.2
Over-current test			X	7.3.3
Measurement of the inherent voltage regulation			X	7.3.4
Measurement of ripple voltage and current			X	7.3.5
Measurement of harmonic currents			X	7.3.6
Power loss determination for assemblies and equipment	X			7.4.1
Temperature rise test	X			7.4.2
Power factor measurement			X	7.4.3
Checking of auxiliary devices	X	X		7.5.1
Checking the properties of the control equipment	X	X		7.5.2
Checking the protective devices	x	x		7.5.3
Immunity test			X	7.6 a)
Radio frequency radiated and conducted disturbances			X	7.6 b)
Measurement of audible noise			X	7.7
Additional test			X	7.7

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6.1 TYPE (DESIGN) TESTS:

- All type (design) tests prescribed in the relevant IEC or equivalent ANSI/IEEE shall be successfully executed
- The manufacturer shall provide the EEHC with the detailed design documentation intended for the type tests including: Bill of Material with components specification and part numbers, wiring diagrams, mechanical drawings, etc.
- In lieu of the actual type (design) tests, certified test reports of type (design) tests performed on an identical unit may be submitted to ...EDC for review and approval during bidding stage.

6.2 ROUTINE (PRODUCTION) TESTS

- ...EDC representatives shall have the attached Appendix 1/1 to be filled by him during the tests to ensure the charger complies with the EEHC specification.
- The manufacturer should furnish a list of tests for review and approval by ...EDC representatives.

6.3 OPTIONAL (SPECIAL) TESTS

- Any additional tests which might result from special requirements shall be specified in Data Schedule and performed as per the applicable standard or manufacturer instructions.

7. MARKING:

- General:
All the components of the charger and enclosure should be marked with marking labels according to the applicable IEC standards and should be fitted on all components and enclosure to indicate its technical data. The charger is marked with mimic diagram to indicate the switch type and its position (i.e.: on / off).
- Nameplate
 - The nameplate material shall be stainless steel or other non-corrodible material, but shall not include plastic and shall be fastened to the equipment by stainless steel screws or rivets. The markings shall be imprinted or etched in black, and shall be non-fading.
 - The charger shall bear a nameplate fixed on the front side of the panel. The plate shall be marked in English with the following minimum information:
 - The words "BATTERY CHARGER"
 - Model Number and Type
 - Manufacturer's Name/Country and Trade Mark
 - Year of Manufacture (month and year)
 - Manufacturer's Serial Number
 - Rated AC Input Voltage
 - Rated AC Input Current
 - Frequency
 - Number of Phases
 - Rated DC Output Voltage
 - Rated DC Full Load Output current

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- The supplier guarantees the locally manufactured charger unit against all defects arising out of faulty design or workmanship, or of defective material for period of (3) years from date of delivery according to a bank letter of guarantee with the value of the final insurance.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.
- Maintenance and repair during guarantee period shall be at site.

9. SUBMITTALS

Vendor shall provide the following with the quotation:

- The charger should be certified by the Egyptian Electricity Holding company.
- A complete copy of the operating instructions.
- A complete copy of the maintenance catalog including all the detailed circuit drawings of the charger and the electronic components, if possible.
- Detailed catalogues with full specifications and drawings shall be attached with the offer.
- All necessary drawings including layout, dimensions, single line diagram of primary circuits should be submitted with the offer.
- Technical catalogues containing the technical data for all offered components of the unit should be submitted in addition to routine maintenance procedure
- Graph for fuse holder with dimension and material of components should be submitted.

10. SPARE PARTS (As delivered)

- Selector switch.
- Digital DC Ammeter.
- Digital AC Voltmeter.
- Digital DC Voltmeter.
- Indication lamps.
- Horn
- One trigger board (Fire angle card) for every set of five chargers.
- Communication module.

11. GUARANTEE TABLE

- The tenderer shall fill-in schedules of the technical particulars and performance data.
- By signing the tender, the tenderer guarantees that the equipment supplied will confirm to technical and performance data show in such schedules.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.

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Sec. ref.	Description	Vendor proposed value
4	DESIGN AND CONSTRUCTION REQUIREMENTS	
4.2	Performance characteristics and ratings:	
	– Rated output voltage of the Charger	 VDC
	– Rated Input voltage	 VAC
	– No. of phases and wires of input supply	
	– Charger Output current rating	A
	– Continuous DC load to be catered to by the charger per load profile	A
	– Float charging current to be catered to by the charger	A
	– Battery Recharging time (max.)	Hrs
	– Adjustable charging range for float and Hi-Rate mode	%
	– Steady state Voltage regulations under specified conditions	%
	– Max. Voltage Transients • with battery disconnected • with battery disconnected	
	– Max. Transient Recovery time to (ms) • Steady state conditions Max. • Full recovery	
	– Ripple Voltage with battery disconnected Max	%
	– AC Voltage fluctuations range	%
	– Load imbalance under parallel operation	%
	– Audible noise level at 1.5 m Max.	 dB
4.3	Control and Protection:	
	– Current limit adjustment range Min	%
	– Power walk-in time to full load	Sec
	– Surge protection provided? Yes\No	
	– Short circuit rating of: • AC Circuit breaker	

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	• DC Circuit breaker	 kA kA
4.4	Construction:	
	– Type of mounting, Floor/wall	
	– Conduit/cable entry (top/bottom)	
	– Any Forced cooling required? Yes\No	
	– Total Weight of panel	 Kg
	– Degree of protection	
	– Overall dimensions of charger panel (mm)	
	• Width	
	• Height	
	• Length	
6.	TESTS	
	- Special tests that shall be required (Please mention)	

I / We guarantee the technical data given above for the offered battery charger.

Signature: ----**Date: ---**

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Appendix 1/1

**FAT for Automatic Battery Charger
24Vd.c. /30A**

Project/Station Name:	
Charger Model:	
Charger Serial Nr.:	
Nr. of Phases:	
Input AC Voltage - 50 Hz:	
Charger Output Voltage:	
Charger Rated Current:	
Battery Type:	
Battery nr. of Cells:	
Battery Capacity:	

FAT Procedure**Visual Inspection**

Ensure that all AC, battery and load MCB's are switched off.

Item	Description	Tests Result	
1	All enclosure and metal parts are earthed after equipment installation and check their continuity with earth terminal	☐ Passed	☐ Not Passed
2	Ensure all additional gland openings and any other openings are closed properly	☐ Passed	☐ Not Passed
3	Panel is properly secured to floor	☐ Passed	☐ Not Passed
4	Distance to walls equipment allows free airflow (30cm minimum from each side)	☐ Passed	☐ Not Passed
5	Ensure free access is available to front door.	☐ Passed	☐ Not Passed
6	Panel's painting is in order	☐ Passed	☐ Not Passed

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7	Check the rating and nameplate details.	☐ Passed	☐ Not Passed
8	Marking and identification of external cables.	☐ Passed	☐ Not Passed
9	Ensure the availability of commissioning power with all protection.	☐ Passed	☐ Not Passed
10	Routine factory test sheet provided	☐ Passed	☐ Not Passed
11	Rectification thyristors and diodes are installed full wave bridge	☐ Passed	☐ Not Passed
12	Correct size of MCB's as per EEHC specification section "4.2".	☐ Passed	☐ Not Passed
13	20A Dropping Diodes Regulating Unit with 2 Stages for float and high-rate Installed	☐ Passed	☐ Not Passed
14	Protection lamps as per EEHC specification section "4.3.9.2".	☐ Passed	☐ Not Passed
15	Modbus Communication Provided	☐ Passed	☐ Not Passed
16	All Parameters are settable as per EEHC specification section "4.3.9.1".	☐ Passed	☐ Not Passed
17	Blocking Diode Installed	☐ Passed	☐ Not Passed
18	Varistor for Ac L- N Installed	☐ Passed	☐ Not Passed
19	Transformer Thermistor Installed	☐ Passed	☐ Not Passed
20	Thyristor Thermistor Installed	☐ Passed	☐ Not Passed
21	Correct size of equipment according to Final Drawing	☐ Passed	☐ Not Passed
22	Duty, Standby Terminals Blocks Installed "option"	☐ Passed	☐ Not Passed
23	All voltage and current readings are provided	☐ Passed	☐ Not Passed
24	Earth bar installed	☐ Passed	☐ Not Passed

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Switch on AC, battery and load MCB's. Start performing each of the tests in the table below:

Item	Test Point	Readings	Min. Range	Max. Range
1	Check on the AC input, line to Neutral	V	187.0 V	253.0 V
2	Floating charge (Set at 28 V)	V	28 V	
	2.1: Battery Voltage	V	25 V	30 V
	2.2: Load Voltage (with 30 amper load)	V	23.5 V	24.5 V
3	Hi-Rate charge (Set at 31V)	V	31.0 V	
	3.1: Battery Voltage	V	28 V	33 V
	3.2: Load Voltage (with 30 Amps. load)	V	23.5 V	24.5 V
4	Charger current limitation 30 A	A	0 A	30 A
5	Battery Initial Current limitation set at 0.2C	A	(0.2C * 0.98) A	(0.2C * 1.02) A
6	Press Lamp test (All lamps are on)		All lamps are on? Yes/No	
7	Load Current for 10 minutes	A	30 A	30 A
	Battery Current for 10 minutes	A	20 A	20 A
8	Ripple test without battery at float mode with 30A load current.	%	-	2% P.P.
9	Operation with phase reversal		Passed? Yes/No	
10	Automatic initial function		Passed? Yes/No	
11	Soft start checks		Passed? Yes/No	
12	HI-RATE time check		Passed? Yes/No	
13	Initial time check		Passed? Yes/No	
14	Input filter and current THD		Passed? Yes/No	
15	Modbus communication providing all signals		Passed? Yes/No	
16	All voltages and current readings are provided		Passed? Yes/No	
17	LVD installed and functioning		Passed? Yes/No	
18	Current Regulation (page 7 section "4.2.4".)			
	18.1 Current limit	A	0 A +2%	30 A +2%
	18.2 Short circuit current	A	3 A	30 A

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Remark:

EEHC Inspector:	Manufacturer:
Name:	Name:
Signature:	Signature:
Date:	Date: